

Hand Gesture Recognition using sEMG Signals Based on CNN

Li Bo, Yang Banghua*, Gao Shouwei*, LinFeng Yan, Haodong Zhuang, Wen Wang*

School of Mechatronic Engineering and Automation, Research Center of Brain Computer Engineering, Shanghai University, Shanghai 200444, China

E-mail: yangbanghua@shu.edu.cn

Abstract: At present, the gesture recognition of surface electromyography (sEMG) is widely used. The traditional machine learning method often has the problem of feature extraction and incomplete information, which leads to low recognition rate. In this paper, a convolution neural network (CNN) based on deep learning is proposed for surface electromyography gesture recognition. the MYO Armband is used to collect the data firstly, then the mean square value and moving average window method are used to obtain the effective signal segment, then filtered, normalized preprocessing, and the data is expanded by sliding window. then the data set is fed into the CNN. after convolution, pooling, gradient descent and dropout layer processing, the classification model of 8 gestures is obtained, and then the model is evaluated with test set data. The experimental results show that the average recognition rate of 8 kinds of gesture movements is 96%, which verifies the feasibility of the method.

Keywords: CNN, surface electromyography, gesture recognition

1 Introduction

A surface electromyographic signal (Surface electromyography signal, sEMG) is a bioelectric signal produced by sensors on the surface of human skin that is accompanied by muscle activity, and it reflects the intention of a specific human body action. It also can reflect the functional state of nerve and muscle in real time in the non-injured state^[1]. How to obtain and recognize the motion pattern and motion intention of human limbs from sEMG has become one of the hot issues in related fields^[2-5].

Early research on gesture recognition based on sEMG mainly focuses on three stages: signal preprocessing, feature extraction and pattern classification, the core of which is feature extraction^[6-7]. To a certain extent, it represents the pattern category in which the signal is located, and the selection of different features will greatly affect the classification accuracy. In previous studies, people have tried to obtain the characteristic types that can maximize the characterization signal, and some achievements have been made^[8-10]. However, it highlights the limitation that the classification effect is too dependent on the quality of features, and some valid information between data will be lost because of feature extraction. During the past few years, the emergence of deep learning has completely changed many fields of machine learning, among which convolution neural network (Convolutional Neural Network, CNN) has gradually become the mainstream method of pattern recognition. To some extent, it solves the problem of information loss caused by feature extraction in traditional machine learning algorithms^[11].

To solve the shortcomings of the previous methods of sEMG gesture recognition, and considering that most of the instruments used for sEMG gesture recognition are very expensive and difficult to popularize, this paper uses portable MYO Armband as the medium of human-computer

interaction. A CNN based deep learning method is proposed to realize the recognition of 8 kinds of gesture movements.

2 CNN Architecture

2.1 CNN Theory

Convolutional neural network is developed on the basis of artificial neural network. Its advantage lies in its characteristics of local receptive field and weight sharing. For a single neuron, only the local perception is needed, and the higher level neurons get more advanced information by synthesizing local information. Weight sharing refers to the fact that the parameters of the feature extracted during the convolution operation are independent of position. Based on the above characteristics, the weight parameters of the network can be greatly reduced and the running speed of the algorithm can be improved. classical CNN include the input layer, the hidden layer and the output layer, where the hidden layer consists of multi-layer neural network, mainly convolution layer and pool layer. the function of convolution layer is similar to feature extraction. The output of the convolutional layer is obtained by convolving the input of the previous layer through some cores, and the pooling layer plays a role in reducing dimensionality and reducing overfitting. Compared with traditional machine learning methods, there is no need to artificially make features. Extract information that can make full use of data.

2.2 CNN Structure

To reduce the number of training parameters, To prevent overfitting of small-scale training samples, In this paper, a three-depth network structure is proposed, As shown in Figure 1, It consists of three convolutional layers, which are 32,64,64 filters, The field of view windows on the filter are 2x2,4x4,2x2,Each convolution layer is followed by a nonlinear activation function (rectified linear unit-ReLU), After the first layer of convolution and the second layer of convolution, a maximum pooling layer is connected, and the pooling window is 2x2.The last convolution layer is followed by a Flatten layer, all the feature parameters are compressed into one-dimensional vectors, then a drop layer is connected, and then a full connection layer of 1024 nodes

*Corresponding Author.

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is connected. The last layer of the neural network is also a fully connected layer, where the number of nodes is equal to the number of gestures, and the SoftMax activation function is used. The network is created and trained based on python 3.7, Keras2.2.5 and Tensorflow2.0 Windows PC platforms.

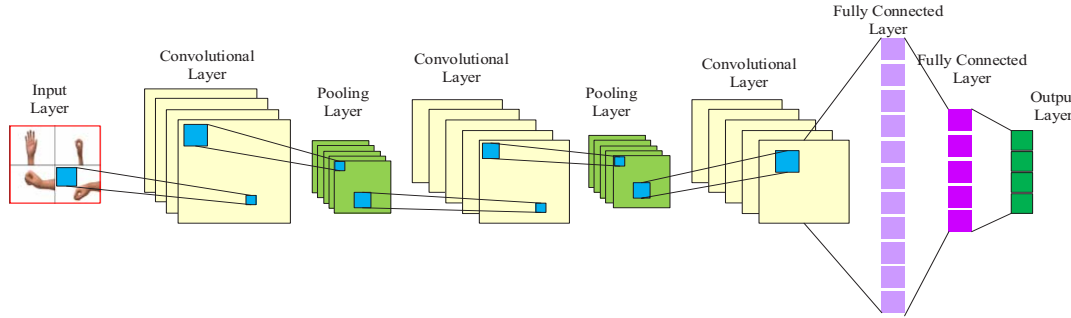


Figure 1 CNN Architecture

3 Acquisition and Processing of Data Sets

3.1 Acquisition Signals

The MYO Armband produced by Thalmic Labs company is used as the acquisition sensor. The MYO Armband has 8 channel surface electromyographic signal sensor, sampling rate 200 Hz, precision 8 bits, easy to wear, simple to use, low cost and so on.

To obtain data samples, this paper designs an experimental paradigm to collect the experimental data of the subjects. After wearing a MYO Armband, the subjects relax and sit in front of the collection PC, and complete the training data collection according to the video prompts on the PC. The process is as follows: First, tell the subjects the next gesture action to be completed through a 2-second guide video, maintain the action for 4 seconds in the task state, and then rest for 2 seconds, and an action experiment ends.



Figure 2 Example of a Paradigm for Offline Training

Each subject completed the 8 actions shown in Figure 3 according to the paradigm designed by the acquisition software. The number of experiments for each action was set to 50 times, and rested for 5 minutes after each group of actions, which can reduce the effect of muscle fatigue on the experiment.

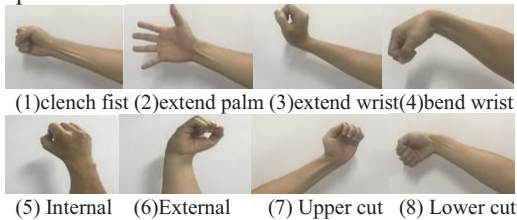


Figure 3 Experimental Action Diagram

3.2 Active Segment Detection

The surface EMG signal collected through the experimental paradigm designed above is a mixture of movement activity segment data and resting state data. If we want to obtain the effective data of the action, we need to

The initialization of the connection weights and parameters between the various layers of the network is completed within the deep learning framework. The neural network has the function of automatic parameter optimization and optimizes the network parameters continuously through the back-propagation mechanism.

further process the original sEMG data. In this paper, the method of combining the mean square value and the moving average window is used to square the original sEMG signal sequence of each channel with a time window of N according to formula (1) to obtain the instantaneous energy:

$$Emg_Energy = \sum_{k=1}^N |x_k|^2 \quad (1)$$

For different gesture movements, only the instantaneous energy value is not enough to determine the start and end point of the action, and then the sum of the energy values of i channels is obtained by formula (2) :

$$Sum_Energy = \sum_{i=1}^i Emg_Energy_i \quad (2)$$

Compare the signal sequence energy sum Sum_Energy obtained in the above formula with the fixed threshold TH . If the signal sequence energy sum before the time stamp is less than the threshold, and the latter is greater than the threshold, then the time stamp is the starting point of the action, and vice versa. This experiment selects 2% of the instantaneous maximum energy of the time series signal as the threshold TH . Figure 4 shows the analysis diagram of the start and end points of the action with the number of channels $i=8$.

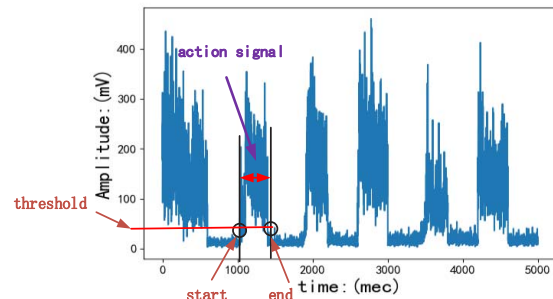


Figure 4 Analysis of Action Start-stop Points

3.3 Preprocess

sEMG is a nonlinear stationary signal [12]. the effective frequency is mainly in the range of 20~350 Hz. In order to reduce the noise interference caused by external or internal factors of the circuit, the six order Butterworth bandpass filter of 20 ~350Hz is used to filter. In addition, in order to

eliminate the adverse effects of single sample data on prediction results, the filtered data is also normalized to keep the data magnitude within $[-1,1]$.

For actual gesture interaction, input delay is an important factor to be considered, and it is mentioned in the literature that the requirement of real-time control response time should not exceed 300 ms^[13]. since the sampling rate of M YO is low, this paper chooses 200 ms time window signal as network input by default. Moreover, in order to meet the requirement of deep learning for data volume, the data needs to be enhanced. This article uses sliding windows to segment the sampled signal. The signal segmentation diagram is shown in Figure 5. 200ms (40 sampling points) is selected as the sliding window length to form a 40x8 two-dimensional matrix information, and at the same time 50ms (10 sampling points) is an incremental window to achieve the expansion of the data set.

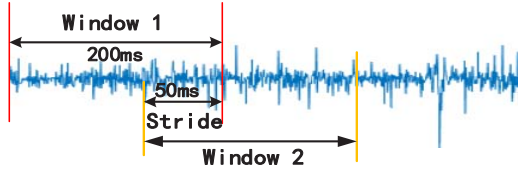


Figure 5 Signal Segmentation Schematic

4 Result and Discussion

By wearing MYO Armband, 10 subjects completed 8 kinds of gesture movements according to the experimental procedure described above, and obtained a series of sample data sets according to the preprocessing method described above. The total samples were divided into training, validation and testing according to the ratio of 7:1:2.

Firstly, the training samples are divided into default time window length (200 ms). The representation information of each trail is a two-dimensional matrix of time window length * channel number. Each training batch uses 200 samples, the number of cycles is selected 50 times, and the verification set verifies the network performance after each iteration. This paper uses cross-entropy as a cost function to calculate the loss value during iteration. in order to reduce the loss, the Adam gradient descent algorithm is used in training because of its adaptability. compared with other gradient descent algorithms (i.e., Adadelta and RMSprop), the exponential decay average of past gradients is retained, the computational efficiency is higher and the convergence is faster. regarding the drop parameter, the default probability parameter is set to 0 by default, that is, after the flatten layer, all neurons are connected to the dense layer in a fully connected manner, but through experimental observations, there will be over-learning, so the drop layer is introduced. It can reduce the connection between some neurons, which can enhance the generalization ability of the model and reduce the occurrence of over-fitting. Manually adjust the parameters through 5 times of cross-validation and a randomly selected verification set, and finally select 0.25 as the parameter of the drop layer.

In order to explore the impact of different signal lengths on the classification performance of the model, the training set samples are divided according to different window lengths. This article uses 200ms by default, that is, the sample length is 40. In addition, the samples are divided into 50, 100, 200, 400, 500. From Table 1 and Figure 6, it can be

found that when the signal length is less than 200, the gesture recognition effect is very good, and the best can reach 98%, but when the signal length is higher than 400, its recognition rate becomes very low. The reason is that the learning algorithm used by the convolutional neural network model is the gradient descent method. When the input data is too long, the learning efficiency is not high and it is easy to fall into the local minimum, making the recognition rate of the experiment poor.

Table 1: Recognition rate of network models with different signal lengths

signal length	50	100	200	400	500
accuracy/%	95.8	97	98.4	88	74

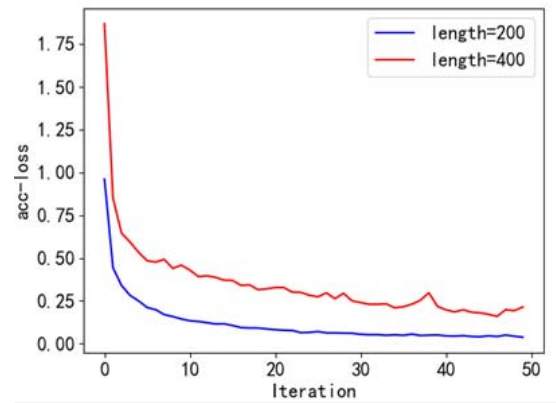


Figure 6 Error curves for different signal lengths

Figure 7 shows the variation trend of training loss function, training accuracy, validation set loss function and accuracy in the network training after the hyperparameter is determined. It can be seen from the figure that the values of training loss function and validation loss function drop quickly when iteration to the second round, indicating that the convergence effect is good. the overall training accuracy maintains an upward trend. Although the test loss function is slightly in the middle, the accuracy of the test is basically maintained at 90%, indicating that the network parameters have a good fitting effect on the training set data.

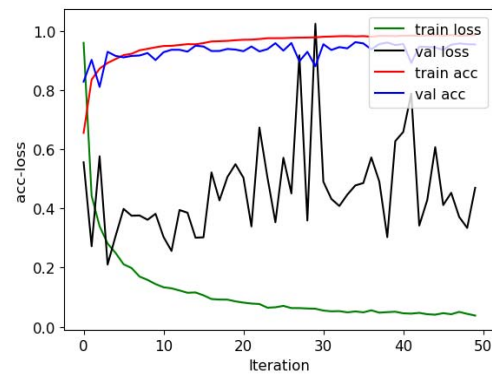


Figure 7 accuracy, loss iterative change diagram

After the network iteration is completed, the classification model is generated, and then the recognition accuracy of the model is checked with the remaining test set. The test set has a total of 1440 samples, 160 samples for each category. The confusion matrix is introduced here, which can obtain the specific recognition situation of each category, and can better analyze the differences between

different gestures. As shown in Figure 8, The overall recognition rate of internal hand gesture recognition corresponding to label 5 is 85%, Sometimes it is recognized as palm extension and Lower cut gestures. The less effective is the external hand gesture, The recognition rate is only 83%, It's easy to be misidentified as an internal gesture, That's because the muscles activity that completes these actions is similar, So signal characterization is easy to be confused. The recognition effect of the other gestures is very good, all reaching more than 98%, and the overall average recognition rate of 8 kinds of gestures has reached 96%.

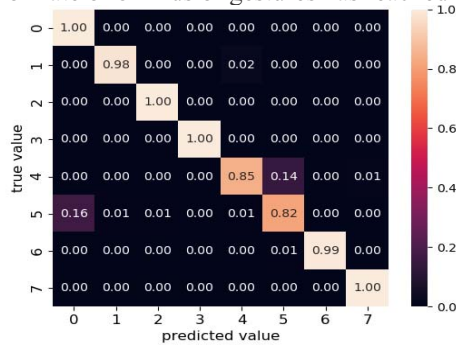


Figure 8 Confusion matrix diagram of test set

To test the robustness of the model trained by the network architecture, we asked three subjects to carry on a set of test experiments on the basis of previous experiments. That is, when wearing the MYO Armband along the vertical direction of the arm 3 cm, and then 8 gesture movements are done 10 experiments. The classification of the model is shown in figure 9. It can be seen from the figure that the overall recognition accuracy will be affected when there is a certain deviation in the electrode wearing position, and the basic decrease is 5 to 10%, which means that the electrode position is one of the factors leading to the poor performance of the classifier. but on the whole, when the electrode position offset is not particularly large, the model of the network structure proposed in this paper has good generalization ability, and the overall average recognition rate is guaranteed to be around 84%.

At the same time, this article is also compared with traditional machine learning methods. This article chooses to use LDA, ANN, RF, and SVM as the standard classifiers, and the sEMG feature uses the feature set with the best performance mentioned in the literature (RMS, TD, HIST, mDWT) [14-15], the recognition rate of different methods under the same test set is shown in Table 2. It can be seen from the table that the recognition rate of the recognition method based on the CNN structure is better than the traditional machine-learning based on feature engineering method.

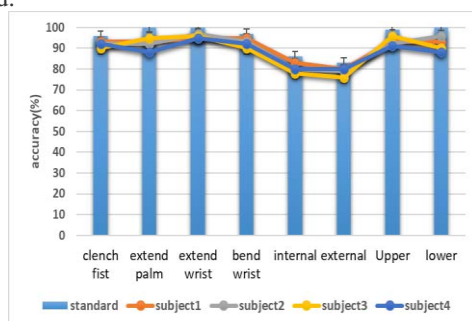


Figure 9 8 Hand recognition accuracy

Table 2: Recognition rates for different classifiers

Classifier	SVM	ANN	RF	LDA	CNN
accuracy(%)	94.5	90	88.9	93.4	95.8

5 Conclusion

This paper proposes a novel gesture recognition method based on CNN. The gesture recognition rate of 8 categories can reach more than 96%, which verifies the feasibility of the method. In the case of slight deviation of electrode position of MYO Armband, the recognition rate can be maintained, which indicates that the classifier is robust. Some new problems have also been found in the experiment. The training data volume based on deep learning is large, and a single individual will spend a lot of time training when using it. In the next work, we need to study how to reduce the complexity of network structure and shorten the training time without reducing the recognition rate.

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